



JULABA

PHD-LEVEL MATHEMATICAL TRADING
SYSTEM

v3.0 | January 2026 | Math-First Architecture



Core Philosophy: Mathematics First

"The main concept is that everything should work with math. AI should use math for scanning. Mistakes are not tolerated here - we are mathematicians."

CRITICAL RULE: Mathematics is the PRIMARY decision maker. AI only validates and refines mathematical decisions - it can NEVER override math with a score ≥ 80 .



Math-First System Architecture



MARKET DATA

15-minute OHLCV candles from Binance



PHD MATHEMATICAL ANALYSIS

12 Quantitative Components | Primary Decision Maker



MATH SCORE CALCULATION

Weighted Score 0-100 | Threshold: 55 (approve), 70 (trade)



AI VALIDATION (Secondary)

Can only CONFIRM, REFINE, or VETO (with restrictions)



COMMAND VALIDATION

Double-layer protection against unauthorized changes



TRADE EXECUTION

Entry with SL/TP based on Kelly Criterion sizing

PhD-Level Mathematical Analysis (12 Components)

Each component is weighted and contributes to a final score of 0-100. Minimum score of 55 required for trade approval.



Trend Strength (SMA)

```
dist = |  
price -  
SMA40| /  
SMA40
```

Measures price distance from 40-period SMA.
Strong trend = larger distance.

Weight: 15%



Volume Confirmation

```
ratio  
=  
volume /  
SMA(volume,  
20)
```

Confirms breakouts with above-average volume.
Ratio ≥ 1.5 = strong.

Weight: 10%



Momentum (RSI)

```
score  
=  
|  
RSI  
-  
50| /  
50
```

Strong momentum when RSI far from neutral 50. Penalizes extremes (>80 , <20).

Weight: 12%



Risk:Reward Ratio

$$R:R = \frac{\text{potential_profit}}{\text{stop_loss}}$$

Minimum 2:1 required. Score scales up to 5:1 for maximum points.

Weight: 12%



Volatility Regime

$$\text{ATR\%} = \frac{\text{ATR}}{\text{price}} \times 100$$

Optimal range: 0.3% - 2.0%. Too low = no opportunity, too high = risky.

Weight: 8%

Hurst Exponent

$$H = \frac{\log(R/S)}{\log(n)}$$

$H > 0.5$ = trending (momentum).
 $H < 0.5$ = mean-reverting.
 $H \approx 0.5$ = random.

Weight: 7%



ADX Strength

$$\text{score} = \min(\text{ADX} / 35, 1.0)$$

$\text{ADX} \geq 25$ = trending.
 $\text{ADX} \geq 35$ = strong trend

(max score)
Weight: 2%

Score (Mean Reversion)

$$Z = \frac{\text{price} - \mu}{\sigma}$$

$|Z| > 2$ indicates potential mean reversion opportunity.



Linear Regression R^2

$$R^2 = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}}$$

Measures trend reliability. $R^2 \geq 0.7$ = strong linear trend.

Weight: 8%

FINAL SCORE CALCULATION

$\text{final_score} = \sum(\text{component_score} \times \text{weight})$

APPROVE if score ≥ 55

OVERRIDE AI if score ≥ 65

PROACTIVE TRADE if score ≥ 70

OVERRIDE AI VETO if score ≥ 80

Mathematical Score Thresholds

Score Range	Classification	Action	AI Role
0 - 54	 REJECTED	No trade allowed	Cannot override
55 - 64	 MARGINAL	AI decides	Primary decision maker
65 - 69	 APPROVED	Trade approved	Can suggest refinements
70 - 79	 STRONG	Proactive trade signal	Validates only
80 - 100	 EXCEPTIONAL	Math overrides AI veto	Cannot veto

Signal Generation Logic



LONG SIGNAL

SMA15
crosses
ABOVE SMA40
OR (price
near
BB_lower AND
RSI < 35)

AND
math_score ≥
55

AND R:R ≥ 2.0

AND ATR% ≥
0.1%



SHORT SIGNAL

SMA15
crosses
BELOW SMA40
OR (price
near
BB_upper AND
RSI > 65)

AND
math_score ≥
55

AND R:R ≥ 2.0

AND ATR% ≥
0.1%



Risk Management

Max
Risk
Per
Trade

1-2%

Stop
Loss

2.0

×

ATR

Take
Profit
1

1.5R

Close

50%

Take
Profit
2

2.5R

Close

30%

Kelly/
2

POSITION SIZE CALCULATION
multiplier

```
kelly = (win_rate × avg_win - loss_rate) / avg_win
position_size = (balance × risk_pct) / stop_distance
risk_pct = min(kelly / 2, 0.02) // Half-Kelly, max  
2%
```



Mathematical Command Validation (Double-Layer)

All settings changes require explicit user intent. AI cannot accidentally modify system parameters.

Layer 1: MATH GUARD (`ai_filter.py`)

Strips unauthorized command blocks from AI responses before they reach execution layer.

Layer 2: Intent Validation (`telegram_bot.py`)

Validates user message against command action before execution.

❌ BLOCKED (Never Execute)

`okay` →
Acknowledgement, not command

`got it` →
Acknowledgement, not command

`what is the risk?`
→ Question, not command

`how does mode work?`
→ Question, not command

`good` →
Acknowledgement, not command

`thanks` →
Acknowledgement, not command

✅ ALLOWED (Execute Command)

`change risk to 2%`
→ Action verb + target + value

`set confidence to 80%` → Action verb + target + value

`switch to BTC` →
Action verb + symbol

`open long` → Trade action + side

`pause the bot` →
Explicit pause request

`resume trading` →
Explicit resume request

Command Type

Required User Intent

Example

set_param

Action verb +
parameter name +
value

"change risk to
1.5%"

open_trade

Trade keywords:
open, buy, long,
short

"open a long
position"

close_trade

Close keywords:
close, exit, sell

"close the position"

Command Type	Required User Intent	Example
switch_symbol	Switch keyword + symbol name	"switch to ETHUSDT"
pause/resume	Explicit pause/ resume/stop/start	"pause the bot"
set_mode	Mode keyword + action verb	"change mode to autonomous"



AI Role Hierarchy

Context	Math Role	AI Role	Who Decides
Signal Filtering	Primary (score calculation)	Secondary (can veto if score < 65)	Math (score ≥ 65 overrides)
Proactive Scanning	Primary (find opportunities)	Validator (confirm/refine/veto)	Math (score ≥ 80 overrides veto)
Chat/ Commands	Intent Validation	Response Generation	Math Guard (strips unauthorized)
Position Sizing	Kelly Criterion calculation	None	Math only
Risk Management	ATR-based stops	None	Math only



Anti-Hallucination Checks

The system continuously validates AI responses against actual system state:

Check Type	What's Validated	Action on Mismatch
Position Hallucination	AI claims position exists vs actual state	Append correction to response
Symbol Hallucination	AI mentions wrong trading symbol	Append correction to response
Balance Hallucination	AI claims wrong balance (>20% off)	Append correction to response
Win Rate Hallucination	AI claims wrong win rate (>15% off)	Append correction to response
Mode/Status Hallucination	AI claims wrong mode or paused state	Append correction to response



Technical Stack

Component	Technology	Purpose
Core Engine	Python 3.9+	Trading logic, math calculations
Exchange	Binance Futures (ccxt)	Market data, order execution
AI Model	Google Gemini 1.5 Flash	Signal validation, chat
Math Libraries	NumPy, SciPy, Pandas	Statistical analysis
Dashboard	Flask + JavaScript	Web interface, control panel
Notifications	Telegram Bot API	Alerts, chat interface
ML Pipeline	Scikit-learn	Pattern learning (future)



JULABA v3.0

PhD-Level Mathematical Trading System

Generated: January 12, 2026 | Math-First Architecture

"Mathematics is the language in which God has written the universe." - Galileo Galilei